

2601-MTL106/MTL2006: TUTORIAL-1

Q1. Let $\Omega = \{0, 1, 2, \dots\}$. Let $\mathcal{F}$ be the collection of subsets of Ω that are either finite or whose complement is finite. Is $\mathcal{F}$ a σ -field? Justify your answer.

Q2. Let $\mathcal{F}$ be a σ -field on Ω and let $B \in \mathcal{F}$. Show that $\mathcal{G} = \{A \cap B : A \in \mathcal{F}\}$ is a σ -field on B . Is $\mathcal{G}$ also a σ -field on Ω ? Justify your answer.

Q3. For $\Omega = \{1, 2, 3, 4\}$, find three different σ -algebras $\mathcal{F}_n$, $n = 1, 2, 3$ such that $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \mathcal{F}_3$. Moreover, define a set function $\mathbb{P} : \mathcal{F}_3 \rightarrow [0, 1]$ such that $(\Omega, \mathcal{F}_3, \mathbb{P})$ becomes a probability space.

Q4. Let $A_r : r \geq 1$ be events such that $\mathbb{P}(A_r) = 1$ for all r . Show that $\mathbb{P}(\cap_{r=1}^{\infty} A_r) = 1$.

Q5. Let $A_1, A_2, \dots, A_N$ be a system of independent events i.e.,

$$\mathbb{P}(\cap_{i=1}^r A_i) = \prod_{i=1}^r \mathbb{P}(A_i), \quad r = 2, 3, \dots, N$$

Assume that $\mathbb{P}(A_n) = \frac{1}{n+1}$, $n = 1, 2, \dots, N$.

- a) Find the probability that exactly one of the A_i 's occur?
- b) Find the probability that at most two A_i 's occur?

Q6. An urn contains balls numbered from 1 to N . One ball is selected uniformly at random.

- i) Find the probability that the number on the selected ball is divisible by 3 or by 4.
- ii) Determine the limiting value of the probability obtained in part (i) as $N \rightarrow \infty$.

Q7. Show that, for any events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A|A \cup B) \geq \mathbb{P}(A|B).$$

Q8. Two factories, Factory (I) and Factory (II), manufacture ballpoint pens. Factory (I) produces twice as many pens each week as Factory (II). Of the pens produced by Factory (I), 20% are defective, while 5% of the pens produced by Factory (II) are defective.

A pen is selected at random from the total weekly production.

- (a) What is the probability that the selected pen is not defective?
- (b) If the selected pen is defective, what is the probability that it was produced by Factory (I)?

Q9. On flights from Delhi to Kolkata, 90% depart on time and arrive on time, 6% depart on time but arrive late, 1% depart late but arrive on time, and 3% depart late and arrive late. What is the probability that, given a flight late, it will arrive on time?

Q10. A biased coin, with probability $\frac{1}{3}$ of head turning up, is tossed repeatedly. Let p_n be the probability that even number of heads has occurred after n tosses (zero is an even number). Show that

$$p_0 = 1, \quad p_n = \frac{1}{3}(1 - p_{n-1}) + \frac{2}{3}p_{n-1}, \quad n \geq 1.$$

Solve this difference equation.

Q11. There are two piles of playing cards. The first pile contains 8 cards: 4 aces, 2 kings, and 2 queens. The second pile also contains 8 cards: 1 ace, 4 kings, and 3 queens.

A card is selected at random from the first pile and placed into the second pile. The second pile is then shuffled, and one card is drawn at random from it. If the card drawn from the second pile is an ace, what is the probability that the card transferred from the first pile was also an ace?

Q12. The coefficients a , b , and c of the quadratic equation $ax^2 + bx + c = 0$ are obtained by rolling a fair six-sided die three times, with the outcomes taken as the values of a , b , and c , respectively. What is the probability that the quadratic equation has two equal real roots?

Q13. Let A_1, A_2 and A_3 be independent events. Prove that

$$\mathbb{P}(A_1 \cup A_2 \cup A_3) = 1 - \prod_{i=1}^3 (1 - \mathbb{P}(A_i)).$$

Q14. Let A and B be independent events. Prove or disprove that A and B^c , as well as A^c and B^c are independent.

Q15. A fair die is rolled five times. Let A_{ij} be the event that the i th and j th rolls show the same number, where $1 \leq i < j \leq 5$. Prove that the events $\{A_{ij} : 1 \leq i < j \leq 5\}$ are pairwise independent but not independent.

Q16. Let $\Omega = \{1, 2, 3, \dots, 7\}$, $\mathcal{F}$ be the power set of Ω , and $\mathbb{P}(A) = \frac{|A|}{7}$ for all $A \in \mathcal{F}$. Show that, if A and B are independent events, then at least one of A and B is either empty set or Ω .

Q17. Two fair six-sided dice are rolled. Show that the event that the sum of the numbers shown is 7 is independent of the number shown on the first die.